

Recurrence of First Afebrile Unprovoked Seizure and Parental Consanguinity: A Hospital-Based Study

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Abstract

Purpose: The aim of the current study was to determine the incidence, clinical characteristics, and risk factors associated with the recurrence of first unprovoked seizure in children. **Methods:** A retrospective, observational study was conducted at King Abdullah University Hospital in Jordan. Children aged from 1 month to 16 years old who attended the hospital between January 2013 to December 2017 were evaluated on the basis of medical records, from the first visit and for a 1-year follow-up period. **Results:** During the study period, a total of 290 cases with first unprovoked seizure were included. The incidence of first unprovoked seizure was 441 cases per 100 000 patient visits to the pediatric clinic. More than half of the cases developed a second attack (55.3%). Children with parental consanguinity were almost 3 times more likely to develop a second attack of seizure compared to those without parental consanguinity (odds ratio [OR] = 2.785, 95% confidence interval [CI] = 1.216-6.378, $P = .015$) and patients who had a history of focal type of seizure were almost twice as likely to develop seizure recurrence (OR = 1.798, 95% CI = 1.013-3.193, $P = .045$). **Conclusions:** The current results showed a high incidence of first unprovoked seizure among children in Jordan. Parental consanguinity and focal seizure were associated with the increased risk of recurrent attack. This finding highlights the need for public education regarding the outcomes of parental consanguinity to improve the patient's quality of life.

Keywords

first unprovoked seizure, children, incidence, parental consanguinity, recurrence

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A first unprovoked seizure is a seizure that is not associated with an obvious precipitating cause.¹ The approach to a child with a first unprovoked seizure is challenging for several reasons, such as difficulty in establishing the diagnosis, the means of investigations, and whether antiepileptic treatment is necessary or appropriate.² Population-based studies have revealed that 25 000-40 000 children per year in the US experienced a first unprovoked seizure.^{1,3} The incidence of first unprovoked seizures was estimated in children to be between 30 and 204 per 100 000 people per year in Sweden; the highest incidence was among the younger children, and the lowest was among the 18-year-olds.²

Recognition of risk factors for recurrence of first unprovoked seizure in children would be a reliable tool for early detection of patients at high risk and provision of early medical interventions (eg, giving antiepileptic medications) and family counseling. The vast majority of studies performed on the risk of recurrence have been carried out in the United States and Europe.^{4,5} Different factors were reported to be related to the

increase in the risk of recurrence of seizure, such as the presence of focal seizure, nocturnal seizures, a history of prior brain injury, a family history of epilepsy, abnormal neurologic examination, abnormal electroencephalogram (EEG) results, abnormal video-EEG, and neuroimaging abnormalities.⁶⁻⁸ In Jordan, a prospective study was conducted in 2004 that investigated the

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risk of recurrence after a first unprovoked seizure in children.⁹ It found that 37% of the children experienced a second attack over the 3-year follow-up period. The risk of recurrence was significantly higher among children with focal seizure (55%) and in those with a positive family history of epilepsy (59%).

Previous studies have shown wide variation in the recurrence rates among those who have experienced a first unprovoked seizure.^{4,10} For example, Maia et al reported that the overall recurrence rate over 5 years was 38%, and that 80% of the recurrent cases had the second seizure within 6 months of the first event.⁴ The patient and the family should be aware of the possibility of seizure recurrence and be involved in any decision to start treatment, or not, after the first attack.¹¹ However, although treatment of the first unprovoked seizure with antiepileptic drugs reduces the risk of a subsequent seizure, this must be balanced against the risk of adverse effects. Importantly, evidence suggests that such treatment does not affect the long-term prognosis of epilepsy.^{12,13} More than a decade ago, a previous study in Jordan investigated the risk factor of seizure recurrence.⁹ However, the current study investigated the incidence of first unprovoked seizure and its clinical characteristics and provides an update on the risk factors associated with recurrence of first unprovoked seizure in pediatric patients.

Methods

Study Design

A retrospective study was conducted in a King Abdullah University Hospital, the main tertiary referral center in the North of Jordan. The study was approved by the Ethics Committee at Jordan University of Science and Technology and was performed in accordance with the Declaration of Helsinki (1964) and its later amendments.

Case Selection

A list of patients with a first unprovoked seizure event from the age of 1 to 16 years who attended the hospital between January 1, 2013, and December 31, 2017, was identified using the hospital database. The diagnosis of first unprovoked seizure in these patients was confirmed by a consultant neurologist using the criteria and classification of the International League Against Epilepsy.¹⁴ The hospital records of these cases were reviewed for 12 months to determine the incidence of recurrence. Recurrence was defined as an unprovoked seizure occurring more than 24 hours after the first seizure. The data were collected for each patient on a structured specific form that included the following variables: date of birth, sex, neurodevelopmental comorbidity (eg, developmental delay, speech/language and learning difficulties, intellectual disability, cerebral palsy, autistic spectrum disorders, attention-deficit hyperactivity disorder [ADHD], other specified psychiatric disorder [tics, depression, and anxiety]), and clinical and medical data including family history, neuroimaging study, EEG, history of febrile seizure, characteristic of seizure episode. Phone calls with the parents were used to check for any missing information. Children who had experienced previous neonatal seizures, febrile seizure, myoclonic seizures, absences, infantile spasms, epileptic encephalopathy, or diagnosed epilepsy were excluded.

Statistical Analysis

Continuous data were presented as median and interquartile range whereas categorical data were expressed as counts and percentage. The incidence rate of first unprovoked seizure was calculated using the total number of patient visits to the clinic over a 5-year period (January 2013–December 2017) and multiplying that number by 100 000. Univariate analysis for factors associated with recurrent attack was examined using *t* test or Mann-Whitney *U* test (continuous variables) and χ^2 test or the Fisher-Exact test as appropriate (categorical variables). All variables with $P \leq .2$ on univariate analysis were included in the multivariate analysis using a binary logistic regression model to determine factors that were independently associated with recurrent attacks. Odds ratio (OR) values and their 95% confidence intervals (95% CIs) were calculated. All tests were 2-sided, and statistical significance was set at $P < .05$. All analysis was carried out using the Statistical Package for Social Sciences (SPSS), version 20.

Results

During the study period, a total of 349 cases with first unprovoked seizure were identified. Fifty-nine cases were excluded for different reasons: febrile seizure only ($n = 36$), juvenile myoclonic epilepsy ($n = 5$), and seizures secondary to presumed trauma ($n = 18$), leaving 290 cases. In the present study, the incidence rate of first unprovoked seizure was 441 cases per 100 000 patients visiting the pediatric clinic.

Table 1 outlines the demographics and clinical data of all included cases. The median age of children when they developed first unprovoked seizure was 5 years, with more than half of them being males (57.9%). Approximately 13% of them had parental consanguinity. Of the 290 first unprovoked seizure cases, more than half (157/284; 55.3%) had a recurrent seizure during the first 12 months of follow-up. Around a quarter of the patients with first unprovoked seizure (74/290; 25.5%) had at least 1 neurodevelopmental comorbidity at presentation. Fewer than half of the patients were admitted to the hospital because of the first unprovoked seizure (119/290; 41%) and 17.6% (51/290) showed abnormal objective examination. Based on the history from parents, more than quarter of the patients (84/285; 29.5%) had a seizure with focal features. More than one-fifth of patients (22.5%, 63/280) had a seizure with onset during sleep, 15.3% (42/275) had a family history of epilepsy, febrile seizure, or cognitive delay, 9.3% (26/281) had prenatal or perinatal complications, and 6.5% (18/276) of cases had a personal history of febrile seizure. Electroencephalography (EEG) was done around the time of the first unprovoked seizure in fewer than half of patients (48.3%; 140/290) and showed abnormal results in 60% (82/136) of those tested. In addition, neuroimaging studies were performed in fewer than half of patients (47.6%; 138/290), and the results were abnormal in 37% (49/132) of cases where results were known.

The results of univariate analysis (Table 2) identified that parental consanguinity ($P = .006$) and type of seizure ($P = .038$) were associated with the risk of recurrent attack. Multivariate analysis adjusted for age and gender revealed that patients with parental consanguinity were almost 3 times as

Table 1. Demographic and Clinical Data of the Study Patients.

Characteristic ^a	
Age at first attack, y ^b	5 (2-9)
Gender	
Female	122 (42.1)
Male	168 (57.9)
Year of data collection	
2013	31 (10.7)
2014	72 (24.8)
2015	74 (25.5)
2016	72 (24.8)
2017	41 (14.1)
Parental consanguinity ^c	
No	240 (86.6)
Yes	37 (13.4)
Admitted to hospital	
No	171 (59)
Yes	119 (41)
Objective examination	
Abnormal	51 (17.6)
Normal	239 (82.4)
Type of seizure/from history ^c	
Focal	84 (29.5)
Generalized	201 (70.5)
Presence of neurodevelopmental comorbidity	
No	216 (74.5)
Yes	74 (25.5)
Seizure with onset during sleep ^c	
No	217 (77.5)
Yes	63 (22.5)
Family history of epilepsy, febrile seizure or cognitive delay ^c	
No	233 (84.7)
Yes	42 (15.3)
Prenatal or perinatal complications ^c	
No	255 (90.7)
Yes	26 (9.3)
Personal history of febrile seizure ^c	
No	258 (93.5)
Yes	18 (6.5)
Developed recurrent attack within 12 mo ^c	
No	127 (44.7)
Yes	157 (55.3)
Performed EEG	
No	150 (51.7)
Yes	140 (48.3)
EEG Results ^c	
Abnormal	82 (60.3)
Normal	54 (39.7)
EEG type ^c	
Focal	39 (52)
Generalized	36 (48)
Neuroimaging study (transfontanelle ultrasound, CT, MRI)	
No	152 (52.4)
Yes	138 (47.6)
Neuroimaging results ^c	
Abnormal	49 (37.1)
Normal	83 (62.9)

Abbreviations: CT, computed tomography; EEG, electroencephalography; FUS, first unprovoked seizure; MRI, magnetic resonance imaging.

^aAll data expressed as n (%) unless otherwise indicated.

^bData are as median (interquartile range).

^cValues were calculated based on the number of patients for whom data were available.

likely to develop recurrent attack compared to those without parental consanguinity (OR=2.785, 95% CI=1.216-6.378, $P = .015$). Additionally, cases who had a history of a focal type of seizure were associated with almost twice the risk of seizure recurrence (OR=1.798, 95% CI= 1.013-3.193, $P = .045$).

Discussion

The present study investigated the incidence rate of first unprovoked seizure and the contributed factors to the recurrence of seizure. The incidence rate of first unprovoked seizure was found to be 441 cases per 100 000 patients visiting the pediatric clinic. More than half of patients developed a recurrent attack after the first episode during a 1-year follow-up period. Children with parental consanguinity and a focal type of seizure were associated with a higher risk of seizure recurrence.

We noted a relatively high incidence of first unprovoked seizure in King Abdullah University Hospital relative to previous studies. A study in Sweden by Andell et al reported an overall incidence of 67 cases of first unprovoked seizure per 100 000 person-years among children, with this varying between 30 and 204 per 100 000 person-years in different age categories.² The findings of Andell and colleagues are not directly comparable to those of the present study, however, because the incidence was calculated differently. In their study, Andell and coworkers calculated the incidence rate by dividing the number of cases of newly diagnosed epilepsy by the official size of the population younger than 19 years. However, the incidence reported in this study was calculated by dividing the total number of patients with first unprovoked seizure by the number of patients who visited the pediatric clinic between 2013-2017, this being because of the absence of a reliable denominator based on the population of children being served by the hospital. Also, the follow-up period in the Andell study was 6 months (although adjusted to a person-years denominator), whereas in our study it was for a period of 12 months.

In the current study, 25% of patients who developed first unprovoked seizure had at least 1 neurodevelopmental problem at the time of presentation. The neurodevelopmental issues were varied and may have been compounded by subsequent epilepsy. However, there was no significant relationship between the presence of neurodevelopmental comorbidity and a recurrent seizure during the 1-year follow-up period in the present study. On the other hand, the earlier study by Andell et al showed that neurodevelopmental comorbidity was more common among those having recurrent seizures.² Additionally, neurodevelopmental comorbidities, including cerebral palsy, autistic spectrum disorders, intellectual disability, ADHD, and learning problems have shown a significant association with childhood epilepsy.^{15,16}

In our study, more than half of the patients developed a recurrent attack during the follow-up period, a finding that is consistent with a study conducted in Japan.¹⁷ In contrast, lower percentages were reported in some previous studies,^{2,4,18-20} including another study from Jordan that was published in 2004.⁹ Of the assessed risk factors in the current study, there

Table 2. Univariable and Multivariate Analysis of Factors Affecting Recurrent Attack.

Characteristics ^a	Univariate analysis			Multivariate analysis	
	No recurrent attack (n = 127)	Recurrent attack (n = 157)	P value	OR (95% CI)	P value
Age at first attack ^b	5 (2-8)	5 (1-9)	.878		
Gender					
Female	53 (41.7)	66 (42)	.96		
Male	74 (58.3)	91 (58)			
Parental consanguinity ^c					
No	114 (92.7)	120 (81.1)	.006	Ref	.015
Yes	9 (7.3)	28 (18.9)		2.785 (1.216-6.378)	
Admitted to hospital					
No	75 (59.1)	91 (58)	.853		
Yes	52 (40.9)	66 (42)			
Objective examination					
Normal	108 (85)	126 (80.3)	.293		
Abnormal	19 (15)	31 (19.7)			
Type of seizure/from history ^c					
Generalized	95 (76.6)	101 (65.2)	.038	Ref	.045
Focal	29 (23.4)	54 (34.8)		1.798 (1.013-3.193)	
Presence of neurodevelopmental comorbidity					
No	98 (77.2)	113 (72)	.320		
Yes	29 (22.8)	44 (28)			
Seizure during sleep ^c					
No	98 (79)	117 (76.5)	.611		
Yes	26 (21)	36 (23.5)			
Family history of epilepsy, febrile seizure, or cognitive delay ^c					
No	109 (87.9)	124 (82.1)	.185		
Yes	15 (12.1)	27 (17.9)			
Prenatal or perinatal complications ^c					
No	112 (90.3)	140 (90.9)	.867		
Yes	12 (9.7)	14 (9.1)			
History of febrile seizure ^c					
No	115 (93.5)	142 (93.4)	.980		
Yes	8 (6.5)	10 (6.6)			
EEG results ^c					
Abnormal	43 (59.7)	36 (60)	0.974		
Normal	29 (40.3)	24 (40)			
EEG type ^c Focal	18 (50)	20 (55.6)	0.637		
Generalized	18 (50)	16 (44.4)			
Results of neuroimaging ^c					
Abnormal	24 (33.8)	24 (40)	0.463		
Normal	47 (66.2)	36 (60)			

Abbreviation: EEG, electroencephalography.

^aAll data are expressed as n (%) unless otherwise indicated.^bData are median (interquartile range).^cValues were calculated based on the number of patients for whom data were available.

was a statistically significant association between parental consanguinity and seizure recurrence. Children with parental consanguinity were 3 times as likely to develop a recurrent attack of seizure as those with unrelated married parents. Consanguinity is a common marital practice in many developing countries, and up to 50% of marriages in some Arab countries are consanguineous.²¹ Jordan has a high prevalence of consanguineous marriages.²² In addition, social traditions, levels of education, and place of residence all play important roles in the rates of consanguineous marriage.^{22,23} According to Jordan

Population and Family Health Surveys (2012), 35% of all marriages were consanguineous in Jordan, and most of those were first-cousin marriages (66%).²² However, the previous study in Jordan found no association between parental consanguinity and recurrent seizures.⁹ This discrepancy might be explained by differences in study design, sampling, and settings. Further prospective longitudinal study of children who had recurrent seizures could assist in identifying the incidence of first unprovoked seizures and of subsequent epilepsy. In particular, it would be useful to know whether the association with

consanguineous marriage is reproducible, and whether this confers a risk for first unprovoked seizure overall in addition to the risk of recurrent seizures given that a first unprovoked seizure has occurred. If this association were found to be reproducible, recognition of patients with a higher risk of recurrence of seizures as a result of consanguinity might suggest the need for early medical intervention, such as possibly administering an antiepileptic medication and providing more comprehensive family counseling and support.

In the current study, the history of focal seizure was associated with a 2-fold increase in the risk of recurrence. Similar results were reported by previous study conducted in Portugal.⁴ An earlier study in Jordan found that focal seizure and positive family history of epilepsy were the most significant risk factors for seizure recurrence.⁹ In contrast, other studies have found no such association.^{4,11} This inconsistency in results could be due to the possibility of semiologic classification bias. Other studies reported associations with seizures during sleep, the presence of neurodevelopmental comorbidity, and EEG and neuroimaging abnormalities^{2,4,17,19,20}; however, we did not observe any of these factors in the current study.

This study has some limitations because of its retrospective design and patients' recruitment from a single health care center. In some cases, it was not possible to verify some clinical data that could have affected the risk of seizure recurrence. In addition, not all needed details were extracted for 12 months of follow-up (eg, level of parents' education) due to poor reporting. However, phone calls were conducted with parents to capture any possible missing data, and all pediatric patients in King Abdullah University Hospital were seen by pediatric neurologists.

Conclusions

Our results suggested that the incidence of first unprovoked seizure among children in Jordan was relatively high. The risk of seizure recurrence was significantly higher in patients with parental consanguinity, and in those with focal seizures. This has the potential to affect the health care budget in countries with low income and limited health care resources such as Jordan. In addition, this finding highlights the need for public education regarding the effects, outcomes, and complications of parental consanguinity and for providing comprehensive family counseling and support. Furthermore, the findings of the current study may contribute to a better understanding of first unprovoked seizure and its management, not just in Jordan but in other Middle Eastern Arab countries, and especially those regions with a high prevalence of parental consanguinity. Further prospective studies that examine this relationship with larger sample sizes are recommended.

Author Contributions

MAM conceived and designed the study and, along with SBH, collected and processed the data. MAM, BA, and AL analyzed and interpreted the data. MAM, BA, and SBH prepared the manuscript, and AL assisted in its revision. The submitted version was approved by all authors.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Ethical Approval

This study was approved by the Institution Review Board (IRB) of King Abdullah University Hospital (KAUH), Jordan University of Science and Technology (JUST).

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